

JOYITA FARUK

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RESEARCH INTERESTS

Computational Biology, AI/Deep Learning, Multi-modal Learning, Graph Neural Networks, Data Analytics

EDUCATION

Bachelor of Science | *Computer Science* | *CGPA 3.54/4.0*
Independent University, Bangladesh

Sep. 2019 – Jul 2024
Dhaka, Bangladesh

RESEARCH

Kidney Abnormality Detection Using Segmentation-Guided Classification on CT Images

Published in: International Conference on Machine Learning and Cybernetics (ICMLC), Miyazaki, Japan.
Pages 414–419. DOI: [10.1109/ICMLC63072.2024.10935113](https://doi.org/10.1109/ICMLC63072.2024.10935113)

Kidney Abnormal Region Localization on 3D CT Images Using ROI-Guided Segmentation

Undergraduate Thesis

WORK EXPERIENCE

Data Scientist, R&D Department
SimpliSolve

Jun 2025 – Present
Texas, US (Remote)

- Researching the latest methodologies to support the development of R&D projects
- Developing state-of-the-art AI models and deploying them using cloud services such as AWS
- Conducting thorough exploratory data analysis
- Writing efficient and robust GPU-compatible Python code
- Leading national- and university-level AI innovation talent hunts across Bangladeshi universities on behalf of the company, designing competition problems and datasets, coordinating logistics, and collaborating with academic and industry partners.

Genomics Intern
Genereveal

Feb 2025 – Jun 2025
Montana, US (Remote)

- Worked on human DNA sequence data, exploring disease associations
- Leveraged Slurm-based HPC for large-scale genomic data processing
- Analyzed genomic data via alignment, variant calling, and tools like BWA, GATK, and SAMtools
- Developed and optimized bash scripts to automate workflows

SKILLS

Machine Learning & AI: PyTorch, Tensorflow, AI Agent Workflows

Cloud & DevOps: AWS, Docker, Terraform, Linux Shell Scripting

Programming Languages: Python, Java, C++

Databases: MySQL, Firebase, Neo4j

Web & Software Development: Flutter

Document Creation: $\text{\LaTeX}$, Markdown, Microsoft Office Suite

Languages: Bengali (Native), English (IELTS Academic 8.5), French (Preparing for A1)

HONORS AND AWARDS

Mitacs Globalink Research Award

January 2025

Competitive award for conducting research internship in Canada; Grant Amount: \$6000 CAD (deferred due to program timing adjustments)

3-Minute Thesis Competition Winner

March 2024

1st Place at the CSE Annual Research Day, Independent University, Bangladesh

Vice Chancellor's List

2024

Awarded for attaining a CGPA of 3.50 or more in the third successive semester at Independent University, Bangladesh

Dean's Merit List

2024, 2022

Awarded for attaining a CGPA of 3.50 or more in the second successive semester at Independent University, Bangladesh

Microprocessor Project Showcase Champion

December 2023

1st Place at IUB Autumn Tech Fest, recognized for technical innovation and presentation

Web Application Project Showcase Winner

July 2023

1st Place at IUB Summer Tech Fest for developing an innovative competition platform prototype

Merit Scholarship

2019–2022

Awarded based on university entrance exam; received full or partial tuition coverage across multiple semesters

WORKSHOPS

Interdisciplinary Computational Biology Workshop 2025

January 2025

3-Day Workshop at BRAC University

Dhaka, Bangladesh

2nd CCDS Workshop on Deep Learning Code Management

December 2024

2-Day Workshop by CCDS, Independent University, Bangladesh

Dhaka, Bangladesh